

Customer Segmentation & Lifetime Value Prediction

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Motivation and Introduction

How can we identify the most valuable customers, and do basic behavioral signals (RFM) provide accurate forecasts of future value? We segment customers and predict 6-month CLV to guide and budget decisions. Once we we cleaned ~0.8M transactions (5,816 customers) from UCI Online Retail II,we model CLV using BG/NBD+GG and XGBoost on R/F/M and tenure features.

Why it matters: Understanding which customers drive value lets companies allocate marketing and retention efforts where they matter most.

Results: XGBoost MAE 172.49, RMSE 643.15, R² 0.843; Pareto shows top 20%≈ 95% of revenue.

Actions: Loyalty for high-value, onboarding for growth, save at-risk, low-cost reactivation.

Data

Source: Downloaded UCI Retail II (UK, 2009-2011, GBP)

<https://archive.ics.uci.edu/dataset/352/online+retail>

Cleaning steps: removed cancellations (“C*”), negatives, missing Customer ID, resulting in:

- 1.1 million transactions before cleaning and 0.8M after
- 5,816 customers

Target/Split: 6-month CLV using a time- based hold-out.

Methodology

Segmentation: We compute Recency (R), Frequency (F), Monterey (M) -RFM features and apply K-means (k=4), visualized with PCA.

CLV: We use BG/NBD + Gamma-Gamma as a probabilistic baseline and train XGBoost on RFM and tenure features.

Innovation: Unlike earlier studies that treated segmentation and CLV modeling as separate tasks, our framework *unifies* them in a continuous feedback loop: unify probabilistic + ML and evaluate Lift@k% in one dashboard.

Experiments & Results

- **Segmentation:** Clear cluster separation in RFM space
- **CLV:** MAE/RMSE/R² + Lift@10%
- **XGBoost:** MAE 172.49, RMSE 643.15, R²=0.843
- **Pareto:** Top 20% → ~95% of revenue

Conclusions

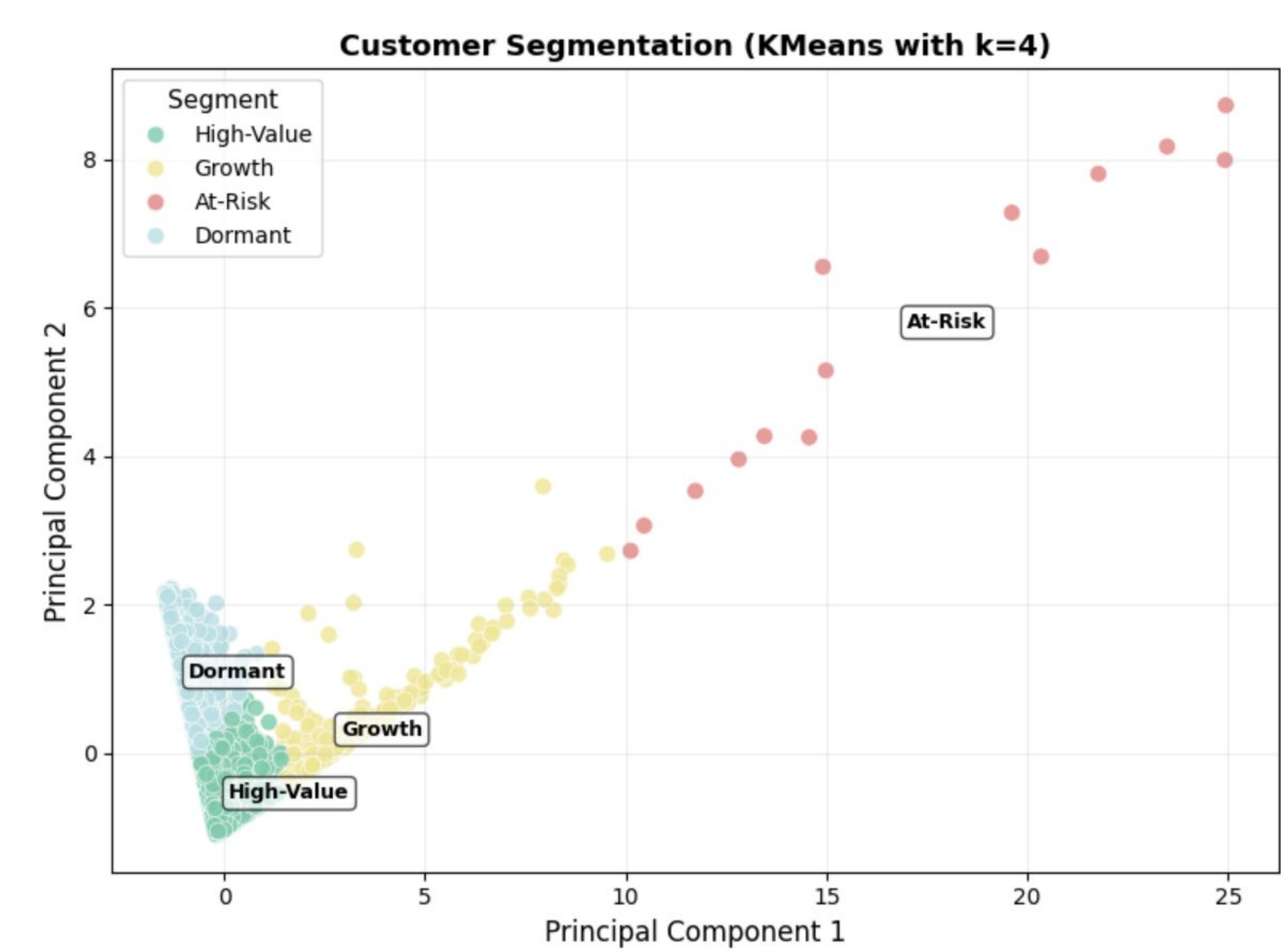
Key Takeaway: Combining behavioral segmentation with CLV modeling provides actionable insights for budgeting, targeting, and retention, with recommendations below:

- * **High-value:** loyalty perks, cross-sell bundles
- * **Growth:** onboarding nudges, cart reminders
- * **At-Risk:** retention offers, service/fulfillment fixes
- * **Dormant:** low-cost reactivation

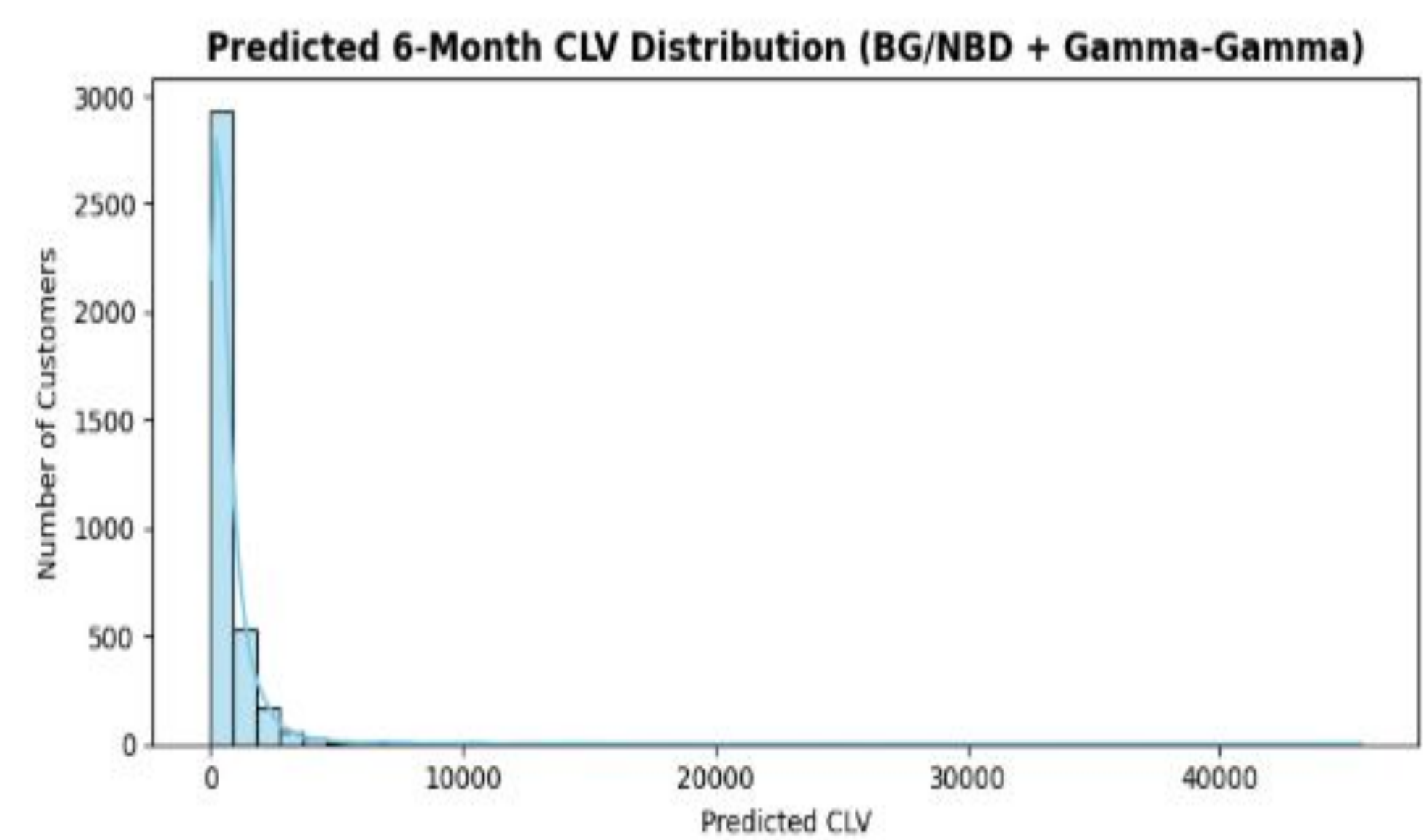
Future Work

Add engagement features and cohort tracking to enhance the performance of the clustering/ML and validate ROI via controlled A/B tests.

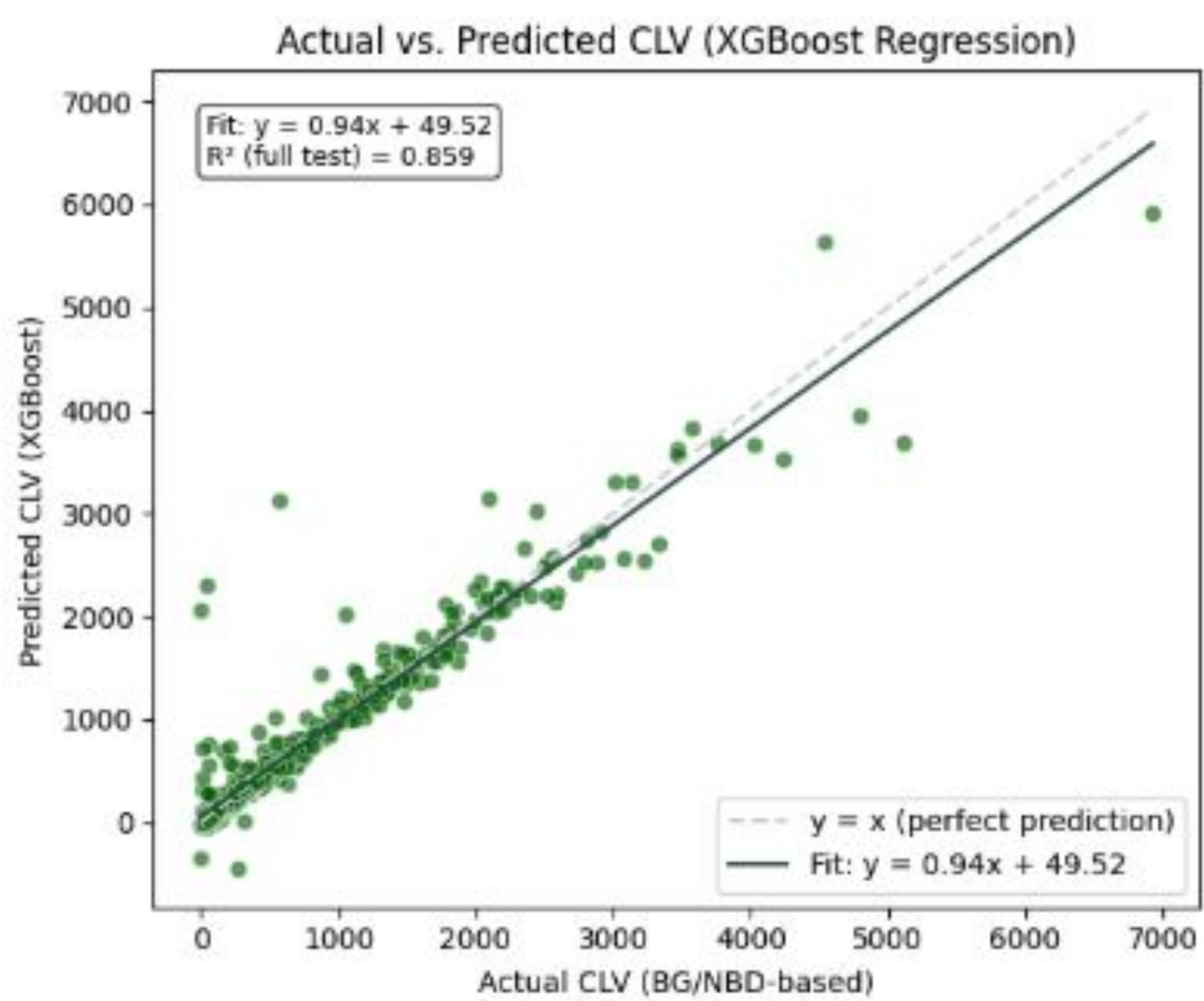
Segmentation (k=4)



CLV Prediction using BG/NBD+GG



XGBoost Regression (Actual vs. Predicted)



Customer Explorer (Recency & Frequency by Segmentation)

